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DEPARTMENT OF DEFENSE
STANDARD PRACTICE

REPAIR WELDING, CLADDING, STRAIGHTENING, AND COLD ROLLING OF MAIN PROPULSION SHAFTING, RUDDER STOCKS, AND DIVING PLANE STOCKS

AMSC N/A FSC 2010

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FOREWORD

1. This standard is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

2. This standard incorporates the Advanced Notice of Change 1 to MIL-STD-2191 distributed by NAVSEA in 2004 via NAVSEA Ltr Ser 05M2/049 dtd 23 March 2004 for updated references, filler metal allowances, welder operator qualifications, clarifications, and other improvements.

3. Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, D.C. 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1. SCOPE

1.1 Scope. This standard contains the minimum requirements for repair welding, clad welding, straightening, and cold rolling of new and refurbished steel main propulsion shafting, rudder stocks, and diving plane stocks for ships and submarines of the U.S. Navy.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-E-21562	-	Electrodes and Rods – Welding, Bare, Nickel Alloy
MIL-DTL-22200/1	-	Electrodes, Welding, Mineral Covered, Iron-Powder, Low-Hydrogen Medium and High Tensile Steel, as Welded or Stress-Relieved Weld Application
MIL-E-22200/3	-	Electrodes, Welding, Covered: Nickel Base Alloy; and Cobalt Base Alloy
MIL-S-23284	-	Steel Forgings, Carbon and Alloy, for Shafts, Sleeves, Propeller Nuts, Couplings, and Stocks (Rudders and Diving Planes)
MIL-E-23765/1	-	Electrodes and Rods – Welding, Bare, Solid and Alloyed Cored, Ordinary Strength and Low Alloy Steel
MIL-E-23765/2	-	Electrodes and Rods – Welding, Bare, Solid, or Alloy Cored; and Fluxes, Low Alloy Steel
MIL-DTL-24403/1	-	Electrodes – Welding, Flux Cored, Ordinary Strength and Low Alloy Steel

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2035	-	Nondestructive Testing Acceptance Criteria

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AQ-GIB-010/248	-	Requirements for Welding and Brazing Procedure and Performance Qualification
S9074-AR-GIB-010/278	-	Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels
S9243-AW-TRS-011/SHAFT	-	Technical Repair Standard for Surface Ships Main Propulsion Shafting Refurbishment Procedures
T9074-AS-GIB-010/271	-	Requirements for Nondestructive Testing Methods
T9074-BC-GIB-010/0200	-	Filler Materials for Critical Applications: Requirements for Flux-Cored Welding Electrodes, Bare Welding Electrodes and Fluxes, and Covered Welding Electrodes for Low-Alloy Steel Applications

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(Copies of these documents are available online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN WELDING SOCIETY (AWS)

AWS A3.0M/A3.0	-	Standard Welding Terms and Definitions, Including Terms for Adhesive Bonding, Brazing, Soldering, Thermal Cutting, and Thermal Spraying
AWS A5.01M/A5.01	-	Welding and Brazing Consumables – Procurement of Filler Metals and Fluxes

(Copies of these documents are available online at www.aws.org.)

ASTM INTERNATIONAL

ASTM E8/E8M	-	Standard Test Methods of Tension Testing of Metallic Materials

(Copies of this document are available online at www.astm.org.)

SAE INTERNATIONAL

SAE AMS2430	-	Shot Peening
SAE AMS2431/1	-	Peening Media (ASR), Cast Steel Shot, Regular Hardness (45 to 52 HRC)
SAE AMS2431/7	-	Peening Media, Ceramic Shot
SAE AMS2590	-	Rotary Flap Peening of Metal Parts
SAE J2277	-	Shot Peening Coverage Determination

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 General. Welding nomenclature and definitions are as stated in AWS A3.0M/A3.0. The following definitions are excerpts from AWS A3.0M/A3.0 and are provided for convenience.

3.1.1 Buildup. A surfacing variation primarily used to achieve required dimensions.

3.1.2 Cladding. A surfacing variation primarily used to improve corrosion or heat resistance.

3.2 Non-standard terms.

3.2.1 Completed weld. Welding is completed when preheat is removed, the material has cooled to ambient temperature, and the weld has been visually accepted and is ready for other nondestructive test (NDT) inspections.

3.2.2 NAVSEA authorized representative. Any Government representative specifically authorized to approve equipment, material, or procedures within the scope of this standard for NAVSEA. These representatives are as follows:

a. For Government shipyards: The delegated representative of the shipyard commander. This includes all applicable areas in the shipyard and items and work contracted by the shipyard.

b. For commercial shipyards: The delegated representative of the Supervisor of Shipbuilding, Conversion and Repair (SUPSHIP) or the Regional Maintenance Center (RMC), as applicable. This includes all applicable areas in the shipyard and applicable items furnished to the shipyard by subcontractors.

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c. For Government purchase items (other than Government shipyards): The delegated representative of the Commanding Officer, Naval Surface Warfare Center, Carderock Division (NSWCCD) and the Commanding Officer, Naval Surface Warfare Center, Philadelphia Division (NSWCPD).

d. A technical representative specifically authorized by NAVSEA.

4. GENERAL REQUIREMENTS

4.1 Repairs.

4.1.1 Major repairs. The following shall be classified as major repairs:

a. Joining two sections of shafting or stocks together.
b. Replacing a flange.
c. Weld buildup of ⅜ inch or greater in the final machined condition.
d. Weld buildup of any thickness that covers more than 25 percent of the shaft or stock surface.
e. Weld repair of cable wrap grooves ½ inch or deeper in the final machined condition.
f. Weld repair of cracks ½ inch or deeper in the final machined condition.
g. Weld repair of cladding within ⅛ inch of the steel base metal.
h. Weld repair of steel beneath cladding.
i. Weld buildup of any thickness that covers 5 percent or more of the surface area for any of the following areas:
    (1) Tapers
    (2) Flanges
    (3) Fillets
    (4) The shaft area in the way of sleeves

4.1.1.1 Welding of major repairs. Welding of repairs classified as major shall be accomplished after extending the weld repair groove 360 degrees around the shaft circumference to minimize distortion and equalize weld stresses. Deviation from this requires NAVSEA approval.

4.1.2 Minor repairs. Repairs shall be considered minor if they involve buildup of keyways and bolt holes or if they do not meet the definition of a major repair per 4.1.1.

4.1.3 Cracks. All cracks found during pre-repair inspection, repair welding, or final inspection shall be documented and reported to NAVSEA.

4.1.4 Cold rolling. Cold rolling may be required after welding, machining, stress relief, or straightening (see 5.8.1.1 and 5.8.1.2).

4.2 Shafting or stock material. This standard shall be followed for repair of all steel shafting and stocks. Repairs to shafting and stocks made of steel materials not listed in table I require NAVSEA approval.

4.2.1 Carbon steels. Steel shafts or stocks manufactured with class 3 or class 4 material in accordance with MIL-S-23284 are considered S-1 group materials per NAVSEA S9074-AR-GIB-010/278.

4.2.2 Alloy steels. Steel shafts or stocks manufactured with class 1 and class 2 material in accordance with MIL-S-23284 are considered S-11F group materials per NAVSEA S9074-AR-GIB-010/278.

4.2.3 Shafting or stocks with no markings or obliterated markings. When the shaft or stock material cannot be determined from visual markings on the shaft or stock, the end of the shaft or stock shall be etched to reveal any obliterated markings. If the etching reveals obliterated markings, the shaft or stock shall be re-marked with the revealed markings in accordance with MIL-S-23284 and the shaft or stock shall be repaired accordingly. If the shaft or stock has no markings or etching reveals no markings, the shaft or stock shall be tested for chemical composition and hardness. A sample shall be removed for chemical composition testing from an area that has not been weld repaired or clad. Five hardness readings shall be taken on areas that have not been cold rolled or welded. The results of the hardness readings and chemical composition testing shall be compared with the heat analysis chemical composition and hardness requirements of MIL-S-23284. If the testing performed meets the requirements of a material class listed in MIL-S-23284, the NAVSEA Authorized Representative may approve repairs to the shaft or stock based on the material class of MIL-S-23284 determined by the testing. The shaft or stock shall be marked on the end with the identified material class in accordance with MIL-S-23284.

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4.2.4 Other steel materials. For other steel materials not specified in table I, the following information shall be determined and submitted to NAVSEA for approval prior to welding, cladding, straightening, or cold rolling:

a. A minimum of five hardness readings taken on areas that have not been cold rolled or welded. Hardness measurements shall be spaced at equal distances around the shaft circumference. Standard Rockwell (B or C scale) or Brinell (standard ball) hardness readings are acceptable. Micro-hardness testing (i.e., diamond pyramid) is not acceptable.

b. When the chemical composition of the shaft or stock cannot be determined from markings, a sample for chemical composition testing shall be removed from an unwelded portion of the shaft or stock. As a minimum, the sample shall be analyzed for carbon (C), manganese (Mn), molybdenum (Mo), nickel (Ni), chromium (Cr), vanadium (V), silicon (Si), sulfur (S), and phosphorus (P). The weight percentages shall be reported to two decimal places (X.XX) for all elements except phosphorus (P) and sulfur (S), which shall be reported to three decimal places (X.XXX).

c. The shaft or stock NAVSEA drawing number.

d. The proposed stress relief processing and temperature.

e. The extent of repair.

f. A description of the defects (i.e., type, size, and location).

g. Any marking found on the shaft.

h. The proposed welding procedure and qualification data.

5. DETAILED REQUIREMENTS

5.1 Welding requirements.

5.1.1 Welding procedure qualification. Procedures for welding and cladding of shafts and stocks shall be qualified to the requirements of NAVSEA S9074-AQ-GIB-010/248 and 5.1.1.1 of this standard. Procedures qualified to earlier revisions of this standard and NAVSEA S9074-AQ-GIB-010/248 where qualification data is available shall not require requalification to this revision. The NAVSEA Authorized Representative may require requalification to this revision when questions arise about the adequacy of the existing procedures to perform welding or cladding to the requirements of this standard.

5.1.1.1 Procedure qualification requirements.

5.1.1.1.1 Welding procedure essential elements. The following shall be considered essential elements in addition to the essential elements specified in NAVSEA S9074-AQ-GIB-010/248, as applicable:

a. Weld sequencing plan for distortion control.

b. Additional methods of distortion control (e.g., peening).

c. Method of measuring distortion during welding.

d. Rotational speed (i.e., rim speed) of the shafting or stock for automatic, mechanized, or semi-automatic welding processes in rolled (i.e., rotated) positions (e.g., 1GR and 2GR).

e. Longitudinal (i.e., axial) speed of the welding torch for automatic or mechanized welding processes when used for helical deposition of weld buildup or cladding.

f. For cladding, bead overlap.

g. Method of stress relief.

h. Shaft or stock rotation speed when stress relief is performed in the rolled horizontal position.

i. Methods of preventing surface scaling during stress relief.

j. Final inspection methods.

5.1.1.1.2 Welding procedure qualification. Welding procedures shall be qualified to the requirements of NAVSEA S9074-AQ-GIB-010/248 as modified below:

a. Radiographic inspection (RT) is not required.

b. Ultrasonic inspection (UT) as specified in 5.3.3 is required.

c. Two transverse tensile specimens and two all weld metal tensile specimens are required.

d. Two macrographic specimens taken transverse to the direction of welding and conforming to the requirements of NAVSEA S9074-AQ-GIB-010/248 are required.

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e. Microhardness surveys covering the weld metal, heat-affected zone, and unaffected base metal shall be performed. The surveys shall traverse the specimen perpendicular to the weld deposit centerline. Both sides of the weld deposition on the same macro face will be surveyed. The maximum linear distance between the microhardness test sites shall be 0.020 inch. The survey shall indicate a relatively uniform hardness level for acceptance.

f. Impact specimens shall be processed as described in NAVSEA S9074-AQ-GIB-010/248. A set of five impact specimens from the weld deposition shall be tested. A set of three impact specimens from the heat-affected zones in both base materials shall be tested. A set of three impact specimens from each base material from an area not affected by welding shall be tested. Impact testing shall be performed to the requirements of NAVSEA S9074-AQ-GIB-010/248. The impact specimen testing will be performed at the temperatures required in the base material specification. Impact specimen test results shall meet or exceed the impact test requirements of the base material specification.

g. Three side bend specimens are required.

5.1.1.1.3 Cladding procedure qualification. Cladding procedures shall be qualified to the requirements of NAVSEA S9074-AQ-GIB-010/248 as modified below:

a. Ultrasonic inspection (UT) of the cladding is required.

b. When the procedure is for applying a corrosion-resistant cladding, chemical analysis of the cladding at the minimum cladding thickness qualified is required. For cladding with Inconel 625, the iron (Fe) content of the cladding shall be less than 9 percent and the molybdenum (Mo) content of the cladding shall be 8 percent or greater.

c. Three tensile tests of the deposited cladding material are required. The tensile test specimens shall be ⅛ inch thick, flat, sub-sized as described in ASTM E8/E8M. The specimens shall be removed normal to the direction of welding. The specimens shall be taken at the minimum cladding thickness qualified. The minimum yield strength at 0.2 percent offset shall be 60 kilopounds per square inch. The other tensile test results shall meet the requirements of the electrode specification.

5.1.1.1.4 Inspection and testing of qualification test assemblies. All NDT and destructive testing of qualification test assemblies shall be accomplished a minimum of 24 hours after the test assembly has reached ambient temperature following stress relief or heat treatment.

5.1.1.1.5 Procedure qualification test assemblies. The welding or cladding procedure qualification test assemblies shall be in accordance with NAVSEA S9074-AQ-GIB-010/248 as modified below:

a. The test assembly shall be cylindrical. Plate product may be used to qualify welding or cladding procedures limited to shaft keyways or flange faces.

b. For welding procedure qualification, the joint designs of figure 1 or figure 2 for shielded metal arc welding (SMAW), figure 3 or figure 4 for gas metal arc welding (GMAW), or figure 5 or figure 6 for submerged arc welding (SAW) shall be used. Qualification using the hollow or solid shaft joint design does not limit the qualification to those shaft types.

c. For a welding procedure qualification to join shafting sections, a 12-inch minimum diameter, 2-inch minimum wall hollow cylinder or a 2-inch deep groove machined into a 12-inch minimum diameter solid cylinder shall be used. These test assemblies will qualify the procedure for all weld deposition thicknesses.

d. For a limited-thickness buildup and weld repair qualification, the thickness qualified will be 1/16 inch to 2T, where T is the weld deposition thickness of the groove weld test assembly. The test assembly thickness shall be sufficient to provide the test samples required by 5.1.1.1.2.

e. Test assemblies welded in the 1G horizontally rolled pipe position qualify 1G rolled and 1G flat position welding only. Test assemblies welded in the 2G vertical fixed pipe position qualify 2G horizontal position and 1G flat position welding only. Test assemblies welded in the 2G vertically rotated pipe position qualify horizontal position welding only when shafting is vertically or horizontally rotated about its long axis (e.g., when welding or cladding flange faces or fillets). Test assemblies welded in the 5G or 6G pipe position will qualify for all position welding. Test assemblies welded in other fixed or rotated positions only qualify for the position welded with an allowed angular deviation of 15 degrees in any plane.

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5.1.1.1.6 Cross qualification.

5.1.1.1.6.1 MIL-S-23284 materials. Welding procedure cross qualification for MIL-S-23284 materials shall be allowed as follows:

a. Welding of class 1 material shall qualify for welding of class 1 and class 2 material.
b. Welding of class 2 material shall qualify for welding of only class 2 material.
c. Welding of class 3 material shall qualify for welding of class 3 and class 4 material.
d. Welding of class 4 material shall qualify for welding of only class 4 material.
e. Welding of class 6 material shall qualify for welding of class 6 and class 5 material.
f. Welding of class 5 material shall qualify for welding of only class 5 material.

5.1.2 Welder performance qualification. To perform welding to this document, welding personnel shall be trained and qualified in accordance with NAVSEA S9074-AQ-GIB-010/248 and the requirements below. Welders shall be instructed on the distortion control processes of the welding procedure.

NOTE: Unless otherwise specified, the table and paragraph references below are those of NAVSEA S9074-AQ-GIB-010/248.

5.1.2.1 Welder qualification. Welder qualification (i.e., for manual and semi-automatic welding) shall conform to NAVSEA S9074-AQ-GIB-010/248; for cladding, UT inspection per 5.3.3 of this document shall also apply.

5.1.2.2 Operator qualification for other than clad welding. Unless otherwise approved by NAVSEA, qualification of welding operators (i.e., for mechanized, automatic, or robotic welding) for groove, buildup, and other welding except clad welding (see 5.1.2.3 below) shall be in accordance with NAVSEA S9074-AQ-GIB-010/248 and the following:

a. For qualification welding in the 1GR position (i.e., shaft in the horizontal position and rotated during welding) or the 2GR position (i.e., shaft in the vertical position and rotated during welding), or any fixed pipe position (e.g., 5G), the test assembly base material form shall be cylindrical (i.e., shaft or pipe product) and the outside diameter shall be 12 inches minimum. Alternatively, operator qualification may be accomplished by welding a plate butt joint in the 1G flat position accompanied by additional training. The written training program shall require utilization of production equipment and setting of procedural essential elements required for welding in the 1G rotated position. The written training program shall be approved by NAVSEA or the NAVSEA Authorized Representative. A pipe test assembly with a wall and weld thickness of at least ⅜ inch may be used in lieu of shaft-type base material forms; this shall qualify thicknesses as specified by Table 7-10 (buildup is not subject to a minimum thickness limit).

b. For production welding of other configurations (e.g., buildup of flange faces and fillets in rotated horizontally oriented shafting), cylindrical or plate base material forms may be used for qualification.

c. Shaft or thick pipe test assemblies may employ a machined (or otherwise prepared) groove joint at least ⅜ inch deep. Where this approach is employed, the test assembly evaluation shall include bend testing, RT, or shear wave UT. RT shall only be performed where acceptance is based on the groove depth, exclusive of additional thickness (e.g., for a ⅜-inch-deep groove in a ⅝-inch-thick pipe test assembly, the ¼-inch extra pipe thickness shall be treated as backing).

d. Where bend testing is used in lieu of RT (or UT) for cylindrical test assemblies, two root and two face bends shall be tested from four locations spaced approximately 90 degrees from each other around the circumference. Where larger thicknesses necessitate, four side bends may be tested in lieu of two root and two face bends. For bend testing of full welded thickness shaft-type test assemblies, one specimen shall include material from the center of the test assembly.

e. If UT is used in lieu of RT, the weld cap shall be machined approximately flush. For longitudinal scanning, the transducer shall cover the full weld width and this coverage shall be specifically recorded on the UT report. The UT inspection zone shall extend at least 3/16 inch below the depth of the machined groove. Otherwise, UT and UT acceptance criteria shall meet the requirements of 4-5.

f. For qualification of carbon and low alloy steels (e.g., S-1, S-11, and all other MIL-S-23284 steels), S-1 material may be used where RT (or UT) is employed for NDT of the test assembly.

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g. For cylindrical test assemblies, qualified positions shall be per those of Table 7-9 for pipe, except that the limits per 5.1.2.4 below shall also apply.

h. The test assembly shall be inspected per Table 7-11, test 4 and as otherwise specified herein (i.e., see 5.1.2.2.b through d above).

5.1.2.3 Operator cladding qualification. Unless otherwise approved by NAVSEA, qualification of welding operators for cladding shall be in accordance with NAVSEA S9074-AQ-GIB-010/248 and the following:

a. In lieu of Figure 7-17A, note 4 diameter requirements, the test assembly diameter shall be within the range specified by the qualified welding procedure. For production welding of configurations other than 1GR, 2GR, 2G, or 5G (e.g., cladding of flange faces and fillets in rotated horizontally oriented shafting), test assembly base material forms may be either cylindrical per Figure 7-17A or plate per Figure 7-17.

b. In lieu of Figure 7-17A, note 5, qualified positions shall be per those of Table 7-9 for pipe except that the limits per 5.1.2.4.b below shall also apply.

c. In addition to the testing specified by Figure 7-17A, note 6 for cylindrical products and Figure 7-17, note 8 for plate, UT in accordance with 5.3.3 of this document shall apply.

d. If composition testing is required by Figures 7-17 or 7-17A, the acceptance criteria for the composition test shall be in accordance with this document (see 5.1.1.1.3.b).

5.1.2.4 Changes requiring operator requalification. In addition to NAVSEA S9074-AQ-GIB-010/248, the following changes shall require requalification of welding operators:

a. For cylindrical test assemblies that are rotated, a change from 1GR to 2GR position welding and vice versa.

b. A change from a rotated position to a fixed position, except that a change to 1G shall not require requalification and a change from 2GR to 2G shall not require requalification.

c. A change from a fixed position to a rotated position unless initial qualification is accomplished by welding a plate butt joint in the 1G flat position accompanied by additional training (see 5.1.2.2.a). A change from 2G to 2GR shall not require requalification.

NOTE: A cylindrical 1GR qualification test and a 2G plate qualification shall qualify all positions in between for rotated weld buildup (or cladding) on flange fillet regions and flange faces in horizontally oriented shafting.

5.1.2.5 Operator qualification records. In addition to NAVSEA S9074-AQ-GIB-010/248, the qualification test record shall include the base material diameter, where applicable.

5.2 Record requirements.

5.2.1 Records. The quality control system shall include the preparation and maintenance of written records. Records shall contain, as a minimum, the following information. (Vendor inspection or mill certificate records will fulfill the requirements of this section for that portion of the required information contained therein.)

a. Welding procedure qualification records in accordance with NAVSEA S9074-AQ-GIB-010/248 and as required herein.

b. Welder and welding operator qualification records in accordance with NAVSEA S9074-AQ-GIB-010/248.

c. NDT procedure qualification records in accordance with NAVSEA T9074-AS-GIB-010/271.

d. NDT personnel qualification records in accordance with NAVSEA T9074-AS-GIB-010/271.

Unless otherwise specified, all required records shall be maintained by the activity and be available to NAVSEA or its authorized representative throughout the life of the contract and for 3 years after delivery. At the expiration of the record retention period, NAVSEA or its authorized representatives shall be given a written notification. Disposition of records shall be as agreed upon by NAVSEA and the contractor.

5.2.1.1 Repairs. Records of repair inspection shall contain the following:

a. Identification of shaft (serial number, ship).

b. Shaft material (chemical analysis as specified in 4.2.3 for materials not listed in 4.2.1 or 4.2.2).

c. Location, size, and orientation of defects.

d. Date and type of original inspection and inspector’s signature.

e. Welding procedure and filler material used.

f. Welder’s identification and signature.

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g. Post weld thermal treatment including charts or recordings of times and temperatures.

h. In-process and final inspections, dates, results, and inspector's identification and signature.

5.2.1.2 Welding electrode control inspection. Records of inspection required by 5.4, including all subparagraphs, shall contain the following:

a. Manufacturer's certification and test results for compliance with the applicable specification.

b. Date of inspection and signature of inspector.

5.2.1.3 NDT. In addition to NDT record requirements in accordance with NAVSEA T9074-AS-GIB-010/271, records of NDT shall include the following:

a. Component description and unique identification.

b. Approved procedure identification.

c. Equipment/material used.

d. Acceptance standard used.

e. Date of inspection.

f. Inspector's signature.

5.3 Inspection requirements.

5.3.1 Visual inspection. Visual inspection shall be performed in accordance with NAVSEA T9074-AS-GIB-010/271. Inspection shall be performed as follows:

a. Areas faired in or prepared for weld repair shall be inspected to ensure defects have been removed and the area blends smoothly into the base material.

b. Each layer of weld deposit shall be inspected. Cracks and lack-of-fusion (including weld rollover) shall be rejected. The final weld surface shall meet the class 1 weld acceptance criteria of MIL-STD-2035.

5.3.2 Magnetic particle and liquid penetrant inspection. Ferromagnetic material shall be magnetic particle inspected. Nonferromagnetic material, or applications where magnetic particle inspection is not practical, shall be liquid penetrant inspected. Inspections shall be performed in accordance with NAVSEA T9074-AS-GIB-010/271. The following areas shall be inspected:

a. Defect removal sites.

b. Completed weld repairs and weld cladding (inspect no sooner than 24 hours after final heat treatment).

c. Machined areas (including areas repaired by fairing in).

d. Areas subject to temperatures above 500 °F (e.g., hot-spot straightening).

e. Areas mechanically worked (e.g., peening, rolling).

Surfaces shall meet requirements of MIL-STD-2035. For welds, class 1 criteria shall apply. Clad areas shall be evaluated to the weld criteria, not the weld overlay requirements. The shaft shall be demagnetized after magnetic particle testing.

5.3.3 Ultrasonic inspection. Inspection shall be performed as follows:

a. Clad surfaces and weld buildup areas that exceed 0.100 inch in thickness after final machining, including flange fillet areas, shall be inspected using the ultrasonic inspection procedure for weld deposited buttering in accordance with NAVSEA S9074-AR-GIB-010/278. If the weld metal and base material are acoustically similar (as defined in NAVSEA T9074-AS-GIB-010/271), it is not necessary to apply weld metal on the calibration block. Inspection shall be performed after post-weld heat treatment. Acceptance criteria shall be in accordance with NAVSEA S9074-AR-GIB-010/278 for weld deposited buttering using the weld thickness as the design material thickness.

b. Other weld repairs (for example, weld repair of cracks) shall be inspected to the ultrasonic weld inspection requirements in NAVSEA T9074-AS-GIB-010/271. Inspection shall be performed no sooner than 24 hours after post-weld heat treatment. Acceptance criteria shall be in accordance with class 1 of MIL-STD-2035. Criteria shall be based on the weld metal thickness or 1 inch, whichever is less. For indications within ¼ inch of any surface, any indication whose amplitude equals or exceeds the disregard level (DRL) shall be rejected.

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5.4 Receipt inspection of welding filler materials.

5.4.1 Responsibility. The receiving activity shall be responsible for verifying that the inspections required in this section are performed and documented. Certified test results for all filler materials, showing chemical and mechanical properties, shall be obtained and reviewed for compliance with the applicable material specification. Level I filler materials received from Naval Supply Systems Command (NAVSUP) by Naval shipyards shall be considered as meeting the chemical and mechanical properties of the filler material, as documented by NAVSUP.

5.4.1.1 Contractor inspection. Where the required inspection is performed by the contractor or by other subcontracted representative prior to receipt, the receipt inspection by the receiving activity shall consist of checking the required documentation against the material received for compliance with the applicable material specification.

5.4.1.2 Receiving activity inspection. Where the required inspection has not been made prior to receipt, the receiving activity shall perform and record the inspections required by the applicable specification.

5.4.2 Nonconforming filler materials. Filler materials which do not comply with the applicable specification shall be isolated and not used for production welding.

5.5 Welding materials and processes.

5.5.1 Approved filler materials and processes. Filler material types and specifications shall be as specified in table I. All carbon and low-alloy steel filler material types shall be conformance inspected in the stress-relieved condition. The use of other processes and materials requires NAVSEA approval.

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TABLE I. Filler materials and specifications. [1/, 8/]

	MIL-S-23284 Class 1	**MIL-S-23284 Class 2**	**MIL-S-23284 Class 3**	**MIL-S-23284 Class 4**	**MIL-S-23284 Class 5**	**MIL-S-23284 Class 6**	**Corrosion resistant cladding [6/]**
Shielded metal arc welding (SMAW)	T9074-BC-GIB-010/0200 MIL-10018-M1	T9074-BC-GIB-010/0200 MIL-10018-M1	T9074-BC-GIB-010/0200 MIL-10018-M1	MIL-DTL-22200/1 MIL-7018-MSR	T9074-BC-GIB-010/0200 MIL-10018-M1	T9074-BC-GIB-010/0200 MIL-12018-M2	MIL-E-22200/3 MIL-1N12
Gas metal arc welding (GMAW)	T9074-BC-GIB-010/0200 MIL-100S-1SR MIL-100S-2SR	T9074-BC-GIB-010/0200 MIL-100S-1SR MIL-100S-2SR	T9074-BC-GIB-010/0200 MIL-100S-1SR MIL-100S-2SR	MIL-E-23765/1 MIL-70S-2	T9074-BC-GIB-010/0200 MIL-100S-1SR MIL-100S-2SR	T9074-BC-GIB-010/0200 MIL-120S-2SR	MIL-E-21562 MIL-EN625
Gas tungsten arc welding (GTAW)	T9074-BC-GIB-010/0200 MIL-100S-1SR MIL-100S-2SR	T9074-BC-GIB-010/0200 MIL-100S-1SR MIL-100S-2SR	T9074-BC-GIB-010/0200 MIL-100S-1SR MIL-100S-2SR	MIL-E-23765/1 MIL-70S-2	T9074-BC-GIB-010/0200 MIL-100-1SR MIL-100S-2SR	T9074-BC-GIB-010/0200 MIL-120S-2SR	MIL-E-21562 MIL-EN625 MIL-RN625
Flux cored arc welding (FCAW)	--	--	MIL-DTL-24403/1 [7/] MIL-80T1-Ni1 MIL-80T1-Ni2	MIL-DTL-24403/1 [7/] MIL-70T-5 MIL-70T-8	--	--	--
Submerged arc welding (SAW)	T9074-BC-GIB-010/0200 MIL-100S-1SR MIL-100S-2SR MIL-100S-1F MIL-100S-2F	T9074-BC-GIB-010/0200 MIL-100S-1SR MIL-100S-2SR MIL-100S-1F MIL-100S-2F	T9074-BC-GIB-010/0200 MIL-100S-1SR MIL-100S-2SR MIL-100S-1F MIL-100S-2F	MIL-E-23765/1 MIL-70S-2 MIL-70S-2F	T9074-BC-GIB-010/0200 MIL-100S-1SR MIL-100S-2SR MIL-100S-1F MIL-100S-2F	T9074-BC-GIB-010/0200 MIL-120S-2SR MIL-120S-2F	MIL-E-21562 MIL-EN625 Neutral flux

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TABLE I. Filler materials and specifications [1/, 8/] - Continued.

STEELS NOT INCLUDED IN MIL-S-23284 OR OBLITERATED SHAFT MARKINGS [2/]							
Maximum tensile strength	**Greater than 95,000 lb/in²**	**Maximum tensile 95,000 lb/in²**	**Maximum tensile 80,000 lb/in²**	**Maximum tensile 75,000 lb/in²**	**Maximum tensile 60,000 lb/in²**	**Maximum tensile 50,000 lb/in²**	**Corrosion resistant cladding [6/]**
Maximum yield strength	**Greater than 65,000 lb/in²**	**Maximum yield 65,000 lb/in²**	**Maximum yield 45,000 lb/in²**	**Maximum yield 40,000 lb/in²**	**Maximum yield 30,000 lb/in²**	----------	
Shielded metal arc welding (SMAW)	[3/]	T9074-BC-GIB-010/0200 MIL-10018-M [4/]	MIL-DTL-22200/1 MIL-8018-C3	MIL-DTL-22200/1 MIL-8018-C3	MIL-DTL-22200/1 MIL-7018-M [5/]	MIL-DTL-22200/1 MIL-7018-M [5/]	MIL-E-22200/3 MIL-1N12
Gas metal arc welding (GMAW)	[3/]	T9074-BC-GIB-010/0200 MIL-100S-1SR MIL-100S-2SR	MIL-E-23765/2 MIL-80S-3	MIL-E-23765/2 MIL-80S-3	MIL-E-23765/1 MIL-70S-2	MIL-E-23765/1 MIL-70S-2	MIL-E-21562 MIL-EN625
Gas tungsten arc welding (GTAW)	[3/]	T9074-BC-GIB-010/0200 MIL-100S-1SR MIL-100S-2SR	MIL-E-23765/2 MIL-80S-3	MIL-E-23765/2 MIL-80S-3	MIL-E-23765/1 MIL-70S-2	MIL-E-23765/1 MIL-70S-2	MIL-E-21562 MIL-EN625 MIL-RN625
Flux cored arc welding (FCAW)	--	--	MIL-DTL-24403/1 [7/] MIL-80T1-Ni1 MIL-80T1-Ni2	MIL-DTL-24403/1 [7/] MIL-70T-5 MIL-70T-8	MIL-DTL-24403/1 [7/] MIL-70T-6 MIL-70T-1	MIL-DTL-24403/1 [7/] MIL-70T-6 MIL-70T-1	--
Submerged arc welding (SAW)	[3/]	T9074-BC-GIB-010/0200 MIL-100S-1SR MIL-100S-1F MIL-100S-2SR MIL-100S-2F	MIL-E-23765/2 MIL-80S-2 MIL-80S-2F	MIL-E-23765/2 MIL-80S-2 MIL-80S-2F	MIL-E-23765/1 MIL-70S-2 MIL-70S-2F	MIL-E-23765/1 MIL-70S-2 MIL-70S-2F	MIL-E-21562 MIL-EN625 Neutral flux

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TABLE I. Filler materials and specifications [1/, 8/] - Continued.

NOTES:

1/ Equivalent filler materials meeting AWS specifications may be substituted for MIL-type filler materials. Acquisition of AWS electrodes shall be in accordance with AWS A5.01M/A5.01, lot classification C4 and S3, for covered and bare electrodes, respectively, and a certified itemized test report in accordance with schedule J is required. Filler materials purchased to AWS specifications shall be as specified by the military specification requirements for conformance testing.

2/ See 4.2 and 4.2.3.

3/ Filler materials selected for these steels shall be as approved on a case basis.

4/ Not used.

5/ Not used.

6/ Corrosion resistant cladding consumables are applicable for all shaft materials for steels included or not included in MIL-S-23284 or shafts with obliterated shaft markings.

7/ Flux cored filler material types conforming to MIL-DTL-24403/1 shall include impact testing in the stress-relieved condition.

8/ MIL-100S-1SR and MIL-100S-1F filler materials may be used for repair welding of MIL-S-23284 class 1, 2, and 3 steels when limited to groove weld or weld buildup of deposit thicknesses 1 inch and less. Use of these filler materials is permitted with a mandatory post-weld hydrogen heat soak to be performed immediately after completion of welding at a temperature of 450 to 500 °F with a minimum hold time of 2 hours for weld deposits less than or equal to ¼ inch thickness, 6 hours for weld deposits greater than ¼ inch but less than or equal to ½ inch, and 12 hours for weld deposits greater than ½ inch but less than or equal to 1 inch, unless otherwise approved by NAVSEA. Production records of post-weld hydrogen heat soaking times and temperatures shall comply with 5.2.1.1.g. The use of MIL-100S-1SR with MIL-100S-2F flux for SAW is permitted without thickness limitation or a mandatory post-weld hydrogen heat soak. Note that, per NAVSEA S9074-AQ-GIB-010/248, hydrogen heat soaks are not procedure level I or II essential elements, but are fabrication document requirements; as such, application and treatment shall be detailed in the welding procedure.

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5.6 Repair requirements.

5.6.1 Approval.

5.6.1.1 Major repairs. Major repairs require the approval of the NAVSEA Authorized Representative. The following information shall be included in the request for major repair:

a. Material specification number, revision number, and alloy class, as in MIL-S-23284-RX-CLX.
b. Heat or forging number (traceability code).
c. Shaft manufacturer's name, trademark, or symbol.
d. Tempering temperature.
e. Stress relief temperature.
f. Drawing or die number.
g. Extent of repair required.
h. Description of defects (type, size, and location).
i. Welding procedure and welding procedure qualification data.
j. NAVSEA serial number.
k. NAVSEA stock number.

5.6.1.2 Minor repairs. Minor repairs may be made at the discretion of the repair activity.

5.6.2 Defect removal. Defective areas shall be removed completely by chipping, grinding, or machining to sound base metal. Grooves or repair areas caused by removal of defects, and surfaces of the shaft adjacent thereto shall be inspected as specified in 5.3, including all subparagraphs.

5.6.3 Repair by fairing. General requirements for repair by fairing for rudder and diving plane stocks shall be as specified in 5.6.3.1 through 5.6.3.3.

5.6.3.1 Minor pitting. Minor pitting, 1/16 inch in depth or less, shall be removed by using a rotary file, wire brush, or similar hand tool. In no case shall the resulting shaft diameter be more than 1/8 inch less than the minimum diameter shown on the applicable drawing.

5.6.3.2 Cracks and severe pitting. Cracks or severe pits 1/8 inch in depth or less for shafts up to and including 10 inches in diameter, or not exceeding 1/4 inch in depth or less for shafts exceeding 10 inches in diameter, may be repaired by grinding and dressing the ground areas to a large faired radius (three times the depth) to avoid stress concentrations. In no case shall the resulting shaft diameter be less than the minimum diameter shown on the applicable drawing by more than 1/4 inch for shafts 10 inches and under in diameter or 1/2 inch for shafts over 10 inches.

5.6.3.3 Additional crack and pitting requirements. The requirements specified in 5.6.3.1 and 5.6.3.2 shall apply to random cracks or pitting and shall be interpreted as allowing grooving to a continuous length of 180 degrees maximum along the shaft circumference. Grooving is not permitted where grooves are 180 degrees diametrically opposite each other. In addition, in any 3-foot length of shaft, no combination of grooves shall additively combine to exceed a length of 180 degrees along the shaft circumference.

5.6.3.4 Submarine shafts. For submarine shafts, fairing shall be in accordance with the applicable Submarine Maintenance Standard as maintained by SUBMEPP.

5.6.3.5 Surface ship shafts. For surface ship shafts, fairing shall be in accordance with NAVSEA S9243-AW-TRS-011/SHAFT.

5.6.4 Weld repair.

5.6.4.1 Preparation for weld repair. The areas to be weld repaired and the adjacent base material within 1 inch of the weld deposition shall be free of oil, grease, and other contaminants that could adversely affect the welding process.

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5.6.4.2 Weld joint design. The weld joint designs of figure 1 or figure 2 for SMAW, figure 3 or figure 4 for GMAW, or figure 5 or figure 6 for SAW shall be used for weld depths greater than 2 inches. Alternative weld joint designs for repairs greater than 2 inches in depth shall require NAVSEA approval. Weld joint geometry shall be selected dependent on the process used to minimize the volume of weld metal deposited for all weld repairs.

5.6.4.3 Welding position. Welding of major repairs in the flat position is preferred. Other positions are allowed as qualified in 5.1.1.1.5.

5.6.4.4 Heat input requirements. The maximum allowed heat input for class 1, class 2, class 5, and class 6 materials of MIL-S-23284 shall be 55,000 joules per inch.

5.6.4.5 Preheat and interpass temperature. The weld joints, weld buildup areas, and cladding areas shall be preheated to the temperatures specified in table II. The maximum welding interpass temperature shall be as specified in table II.

TABLE II. Preheat and interpass temperature requirements.

MIL-S-23284 Shaft Material Class	Minimum Preheat-Maximum Interpass Temperature (°F)
1	400 min – 500 max
2	400 min – 500 max
3	400 min – 500 max
4	275 min – 400 max
5	275 min – 400 max
6	275 min – 400 max

5.6.4.6 Repair of cladding. The classification of weld repairs to corrosion-resistant cladding shall be in accordance with 4.1. Inspection of repaired clad areas shall be per 5.3.

5.6.4.6.1 Major repair of cladding. Where a cladding thickness of less than ⅛ inch remains, and for all other major repairs (for example, extensive weld buildup), the cladding shall be repaired using a welding procedure qualified in accordance with this document.

5.6.4.6.2 Minor repair of cladding. Where a cladding thickness of ⅛ inch and greater remains, the cladding may be repaired using a procedure qualified in accordance with NAVSEA S9074-AQ-GIB-010/248 to the requirements of NAVSEA S9074-AR-GIB-010/278 for class M-1, category A. The welding procedure used shall not exceed a heat input of 55,000 joules per inch. Alternatively, a welding procedure qualified in accordance with this document may be used.

5.6.4.6.3 Repair of steel beneath cladding. Repair of steel defects beneath the cladding layer, or cladding defects that progress into the steel, shall be classified as major repairs (see 4.1.1.h.). The affected area shall be stripped of the cladding material, including the intermixed dissimilar metal layer, a minimum of 1 inch from the defect excavation site. Complete removal of the intermixed dissimilar metal layer shall be verified with an acid etchant that has been demonstrated to distinguish the materials prior to repair welding of the steel. The reduction in steel shaft diameter within the excavation site shall be reported to NAVSEA for acceptance. The intermixed dissimilar metal layer shall be profiled with a 5-to-1 taper out of the excavation. The steel and cladding shall be repaired and inspected in accordance with this document (see 5.3). A single post-weld stress relief of the repaired area (i.e., both steel and cladding) is acceptable; the deposit thickness shall be based on the combined steel groove repair depth and cladding layer thickness.

5.6.4.7 Weld sequencing. Welding shall be sequenced to minimize distortion.

5.6.4.8 Post-weld stress relief. Shafts and stocks shall be stress relieved after welding.

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5.6.4.8.1 Stress relief position. Shafts or stocks shall be stress relieved in the vertical or horizontal position when possible. When the shaft or stock is stress relieved in the horizontal position, the shaft or stock shall be rotated continuously or supported in a manner that will maintain the shaft straightness, within drawing tolerances, while the temperature of the shaft or stock is over 500 °F.

5.6.4.8.2 Heating methods. Furnace heating is preferred for post-weld stress relief of shafting or stocks. Where furnace heating is not practicable, local heating with resistance or induction heaters is acceptable. When local heating is used, the required stress relief temperature shall be maintained in an area that includes the weld repair or cladding and extends a distance equal to at least twice the weld thickness beyond the toe(s) of the weld repair. Other heating methods require NAVSEA approval.

5.6.4.8.3 Stress relief temperature. Shafting and stock stress relief temperatures shall be as follows:

a. If the tempering temperature is marked on the shaft or stock, as required by MIL-S-23284, the shaft or stock shall be stress relieved at 75±25 °F below the marked tempering temperature. The stress relief temperature for MIL-S-23284 class 1 through 4 steels shall be not less than 1,050 °F to avoid embrittlement of the materials.

b. If the shaft or stock is not marked with the tempering temperature, but is marked with manufacturer identification, the manufacturer shall be contacted to determine the tempering temperature. The shaft or stock shall be stress relieved at 75±25 °F below the tempering temperature supplied by the manufacturer. The tempering temperature supplied by the manufacturer shall be marked on the shaft or stock as directed by MIL-S-23284. The stress relief temperature for MIL-S-23284 class 1 through 4 steels shall be not less than 1,050 °F to avoid embrittlement of the materials.

c. If the shaft or stock has no markings but has been determined by the testing in 4.2.3 to be a MIL-S-23284 material, or the manufacturer does not have the tempering records, the temperatures in table III shall be used.

TABLE III. Post-weld stress relief temperature requirements.

MIL-S-23284 Shaft Material Class	Post-Weld Stress Relief Temperature (°F)
1	1,100±25
2	1,125±25
3	1,175±25
4	1,175±25
5	1,050±25
6	1,050±25

5.6.4.8.4 Stress relief holding time. The holding time at stress relief temperature shall be 2 hours per inch of weld deposited thickness. The minimum hold time at stress relief temperature for welds under 1 inch thickness shall be 2 hours.

5.6.4.8.5 Heating and cooling rates. The maximum rate of heating and cooling above 450 °F shall be 150 °F per hour. The shaft or stock may be cooled in still air below 450 °F. Cooling and heating rates shall be monitored by thermocouples attached to the shaft or stock.

5.6.4.8.6 Maximum furnace charging temperature. The maximum furnace charging temperature for stress relief of shafts or stocks shall be 500 °F.

5.6.4.8.7 Temperature monitoring during stress relief. Autographic documenting equipment (analog or digital) shall be used to document stress relief temperature versus time at all temperatures above 450 °F. A minimum of three thermocouples shall be used to ensure and document a controlled uniform stress relief of the shaft or stock. One thermocouple shall monitor the nominal center of the stress relief area. The other two thermocouples shall monitor the temperature a nominal 3 inches from the outer opposing edges of the stress relief area. The thermocouples shall be distributed uniformly about the circumference of the shaft when the shaft is not rotated during stress relief (i.e., a nominal 120 degrees between thermocouples if three thermocouples are used).

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Examples:

a. For a 360-degree groove weld repair, one thermocouple would monitor the temperature at the nominal center of the weld and two would monitor the temperature on either side of the weld 3 inches from the toes of the weld.

b. For a total taper weld buildup, one thermocouple would monitor the temperature at the nominal longitudinal center of the weld deposition. The second would monitor the temperature at the center of the shaft end. The third would monitor the temperature 3 inches from the toe of the weld deposition longitudinally up the shaft.

More than three thermocouples may be required to ensure and document that a uniform stress relief is accomplished. The number of additional thermocouples is dependent on the size of area being stress relieved and the stress relief set-up used.

5.6.4.8.8 Temperature differential during stress relief. The maximum allowed temperature differential during the hold portion of the stress relief shall be a maximum of 25 °F between any attached thermocouples to ensure a uniform stress relief. The maximum allowed temperature differential during the heating and cooling portions of the stress relief shall be a maximum of 75 °F for MIL-S-23284 classes 1, 2, 5, and 6 and a maximum of 50 °F for MIL-S-23284 classes 3 and 4.

5.6.4.8.9 Hardness testing for stress relief. When the shaft or stock tempering temperature information is not available, hardness surveys shall be performed prior to and after stress relief. Prior to stress relief, a minimum of five hardness readings shall be taken in base metal areas of the shaft or stock that have not been welded, shot peened, or cold rolled. After stress relief, a minimum of five hardness readings shall be taken in base metal areas subjected to the full stress relief cycle (i.e., base metal areas subjected to the required stress relief temperature [see 5.6.4.8.2]), avoiding the heat affected zones. Standard Rockwell (B or C scale) or Brinell (standard ball) hardness readings are acceptable. Micro hardness testing (i.e., diamond pyramid) is not acceptable. The hardness readings of the base metal areas that have been stress relieved shall be compared to the hardness readings of base metal areas taken prior to stress relief. The compared hardness values of the two sets of hardness readings shall be nominally equivalent to indicate a satisfactory stress relief. If the hardness levels in the stress relief area are less than the hardness values in the unaffected base metal, but are above the minimum hardness values specified in MIL-S-23284, the stress relief is considered acceptable. If the hardness values are less than or greater than the values listed for the class of material in MIL-S-23284, NAVSEA shall be notified for disposition.

5.7 Straightening.

5.7.1 Shaft material. This standard shall be followed for straightening of all steel shafts. Straightening of shafts not listed in 4.2.1 or 4.2.2, or in cladded areas, requires NAVSEA approval prior to straightening. This section is specifically applicable to shafting material listed in MIL-S-23284 (see 4.2.1 and 4.2.2). The method of straightening shall depend on available facilities, experience of operating personnel, processing condition of shafting, whether new or used shafting is involved, and the extent of straightening required.

5.7.1.1 New shafts. When new shafting requires straightening after the final heat treatment, the straightening process shall be followed by a stress relief performed at a minimum of 50 °F below the final tempering temperature.

5.7.1.2 Used shafts. Except as specified in 5.7.1, used shafting may be straightened. The straightening process shall be followed by a stress relief performed at a minimum of 50 °F below the final tempering temperature.

5.7.2 Procedure.

5.7.2.1 Determine location, magnitude, and type of bend. Prior to initiating straightening processes, the location, magnitude, and type of bend shall be determined. With the entire shaft supported on centers in a lathe, conduct a dial indicator survey of shaft run-out at stations spaced every few feet of shaft length to determine the magnitude and location of the bend. At each station, mark the position of maximum run-out on the periphery of the shaft. The marks on the shaft periphery will be in a straight line if the shaft contains a simple bend and will spiral part way around the shaft if the shaft is both bent and twisted. The maximum run-out shall be plotted on an exaggerated scale to locate the bend center and determine the extent of straightening required.

5.7.2.2 Methods.

5.7.2.2.1 Hot straightening. These methods are intended for use during manufacture of new shafts where the effects of scaling, local distortion due to pressing, and so forth, can be removed in subsequent machining. Other applications require NAVSEA approval.

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5.7.2.2.1.1 During tempering. After the shaft has been held at the tempering temperature for the required time, the shaft shall be removed from the furnace and straightened under a press. The shaft shall be stress relieved after straightening (see 5.7.2.2.1.3).

5.7.2.2.1.2 After tempering. The shaft shall be charged into the furnace at 500 °F or lower, raised to a temperature 50 to 100 °F below the tempering temperature, equalized for 1 hour per inch of thickness, then removed and hot straightened under a press. The shaft shall be stress relieved after hot straightening (see 5.7.2.2.1.3).

5.7.2.2.1.3 Stress relief. After hot straightening, the shaft shall be immediately stress relieved at a temperature at least 50 °F below the tempering temperature but not less than 1,050 °F for 2 hours per inch of maximum thickness, but in no case for less than 2 hours. Stress relief shall be followed by furnace cooling to 600 °F under uniform conditions at a rate not greater than 500 °F per hour divided by the maximum thickness of the part in inches, but in no case greater than 500 °F per hour.

5.7.2.2.1.4 Cold rolling. Cold rolling of the shaft shall be performed after hot straightening and stress-relieving.

5.7.2.2.2 Localized heating (hot spotting). The hot spotting method is intended for straightening used shafting only. Hot spotting involves the iterative use of alternate heating and cooling cycles and provides a means of straightening a section of shafting aboard ship with a minimum of machinery and equipment. Whereas this method offers distinct advantages, it shall be employed with caution so that the shaft is not cracked or damaged.

5.7.2.2.2.1 Hot spotting of steel. For all shafting made of MIL-S-23284 material, the hot spotting temperature shall not exceed the post-weld stress relief temperature of the steel shaft, as determined by 5.6.4.8.3. Magnetic particle testing (as specified in 5.3.2) shall be performed in the affected area(s) after hot spotting is completed and the shaft has cooled to ambient temperature. Previously shot peened or cold rolled areas of the shaft that have exceeded 475 °F as a result of hot spotting shall be re-shot peened or cold rolled after straightening and inspection, in accordance with this document.

5.7.2.2.2.2 Hot spotting of cladding. Liquid penetrant testing (as specified in 5.3.2) shall be performed in the affected clad area(s) after hot spotting is completed and the shaft has cooled to ambient temperature. Previously shot peened areas of the clad shaft that have exceeded 1,000 °F as a result of hot spotting shall be re-shot peened after straightening and inspection, in accordance with this document.

5.7.2.2.2.3 Hot spotting procedure. The following hot spotting procedure shall be used:

a. Determine the location and magnitude of the bend in the shaft as specified in 5.7.2.1.

b. Position the bent section of the shaft with the bend on top (knee-up).

c. Locate and monitor dial indicators as specified in 5.7.2.2.2.4.

d. Apply a concentrated local heat in a circular fashion over a diameter of about 2 to 3 inches at or near the top of the bend. As the metal expands with heat, the bend will increase. The expanding metal will be compressed by the restraining effects of the rest of the shaft. Compressive stresses will be increased with heat input until compressive yielding takes place. When cooled, tensile stresses will develop in the area in which yielding occurred. Tensile stresses will act to oppose the bend and partial straightening will occur. One application may be adequate for small bends while larger bends may require a series of successive spots spread over an area near the maximum run-out.

e. The shaft shall be allowed to cool between each spot and the run-out shall be plotted as specified in 5.7.2.1. This allows evaluation of the effect of the previous hot spot and helps to locate where the next spot should be applied. Post repair stress relief shall not be performed after hot spotting. Cold rolling (see 5.8.1.2) may not be required.

5.7.2.2.2.4 Dial indicators. Before applying local heat, four dial indicators shall be located on the shaft. Two dial indicators shall be placed on the top of the shaft each equidistant on each side of the bent region. Another dial indicator shall be placed 90 degrees from each of the first two. The dial indicators shall be located as far as practical from the area to be hot spotted, but need not be more than 10 feet distant. Prior to application of heat, the dial indicators shall be set to measure distortion in either direction. After hot spotting is completed and the shaft has cooled to ambient temperature, total indicated run out in the shaft shall be a few thousandths of an inch, opposite to the original run-out.

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5.7.2.2.3 Cold straightening. The methods specified in 5.7.2.2.3.1 through 5.7.2.2.4 are intended for use during final machining operations or where it is not feasible to hot straighten. They may be used on used shafting, and on new shafting with NAVSEA Authorized Representative approval.

5.7.2.2.3.1 Cold peening. The location and magnitude of the bend in the shaft shall be determined as specified in 5.7.2.1. The shaft shall be supported approximately 1 to 2 feet on each side of the bend center and positioned so that the high spot of the bend is downward (at 6 o'clock). The shaft shall be peened on the top with sledge hammers, starting at a point over the center of the bend and continued over the shaft in both directions as far as the supports. Peening will raise the bent portion of the shaft to a common axis with the ends. After machining as specified in 5.7.2.2.3.1.1, additional straightening shall utilize protective copper or steel plates, of sufficient thickness to prevent marring of the shaft surface, placed on the shaft surface where blows are to be struck.

5.7.2.2.3.1.1 Machining or grinding after cold straightening. When the shaft is within straightness requirements of the applicable shafting drawing, the shaft shall be machined or ground to remove any inadvertent peening marks, but in no case shall the shaft diameter be reduced by more than as specified in 5.6.3.1. If shafts are machined or ground in regions which require cold rolling in accordance with the applicable shafting drawing, shafts shall be cold rolled after machining. After machining, shafts shall be checked for straightness as specified in 5.7.2.1 and re-straightened if necessary.

5.7.2.2.3.2 Vibration support. The shaft should be supported for straightening by vibration as described below (see figure 7). Methods used to support the shaft need not be limited to those specified as follows:

a. Fasten one end of the shaft to the face plate of a lathe or other device. Suspend the shaft at the center of bend from an overhead crane with the maximum run-out facing down. Allow the other end to hang free.

b. Suspend the shaft from an overhead crane with the maximum run-out facing down and both ends hanging free.

c. Support the shaft, with the run-out facing down, on blocks on either side of the center of bend and approximately 3 feet apart.

d. Support the shaft in saddles on either side of the center of bend with the run-out facing down, and the point of maximum run-out supported with a jack.

5.7.2.2.3.2.1 Vibratory forces. After the shaft has been supported properly, it shall be vibrated (violently) by striking it on top at both ends with a heavy machinist's hammer or a light sledge. The combination of vibratory forces, and bending moments introduced by selected support points will cause the shaft to return to the original neutral axis. Protective plates shall be required as specified in 5.7.2.2.3.1. Machining or grinding shall be as specified in 5.7.2.2.3.1.1. If applicable, re-cold rolling of the shaft shall be performed after machining or grinding (see 5.7.2.2.3.1.1).

5.7.2.2.4 Other straightening methods. For alloy steels (see 4.2.2), straightening methods other than those specified herein require NAVSEA approval. For carbon steels (see 4.2.1), methods require review and approval by the NAVSEA Authorized Representative prior to straightening.

5.8 Cold rolling.

5.8.1 General requirements. This section is applicable to shafting materials listed in 4.2.1 and 4.2.2 and shall be used as a guide for cold rolling of all steel shafting.

5.8.1.1 New shafts. When specified on applicable drawings, cold rolling of propulsion shafting shall be performed on the after end of new propulsion shaft sections. Cold rolling of new shaft sections shall be accomplished prior to cutting propeller keyways or installation of after strut sleeves.

5.8.1.2 Refurbished shafts. For refurbished shafts, cold rolling shall be performed on previously cold rolled surfaces which have been welded, hot straightened, and machined or ground, stress relieved, or otherwise thermally treated. If possible, cold rolling of refurbished sections shall be accomplished prior to installation of after strut sleeves. Dummy keys (see 5.8.2.5) shall be fit to tapers prior to cold rolling. Cold rolling need not be accomplished on shafting that is straightened aboard ship.

5.8.2 Procedure.

5.8.2.1 Finish requirements. The area of the propeller shaft to be cold rolled shall have a 125 microinch root mean square or finer finish before rolling.

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5.8.2.2 Roller sizes. Cold rolling shall be accomplished using a 9-inch diameter hardening roller with a 1½ inch crown radius bearing on the shaft. The hardening roller shall be opposed on the opposite side of the shaft by a 9-inch diameter smoothing roller with a 4-inch crown radius.

5.8.2.3 Rotation and feed. The shaft shall be rotated with a surface speed of about 60 feet per minute and shall be moved through the rollers at a rate of feed of about 1/16 inch for each revolution of the shaft.

5.8.2.4 Shafts.

5.8.2.4.1 New shafts. Cold rolling of new propeller shaft sections shall be continuous from a point 12 inches forward of the forward end of the after shaft sleeve to a point 2 inches forward of the after end of the taper. One pass shall be made over the area to be rolled.

5.8.2.4.2 Repaired shafts. For existing shaft sections where it is necessary to reroll the shaft sections (see 5.8.1.2) rolling shall end 12 inches on either side of the area affected by the operation listed in 5.8.1.2.

5.8.2.5 Rolling tapered sections of existing shaft. When rerolling of an existing tapered section is required, dummy keys shall be fitted to fill keyways. The spooned out area of the forward end of the keyway shall be filled with weld metal and stress relieved prior to rolling to provide a continuous surface for the rollers to bear upon. The welding procedure shall be as specified in 5.1.1, including all subparagraphs therein. The after end of the dummy key shall also be welded in place as necessary to prevent movement during rolling. The keyway sled runner end shall be re-machined and faired to shaft diameter when the dummy key is removed after rolling.

5.8.2.6 Lubrication. The area to be cold rolled shall be coated with an extreme pressure lubricant during rolling.

5.8.2.7 Roller loads. Roller loads shall be as shown in table IV.

TABLE IV. Roller loads. 1/

Yield Point (lb/in^2) (0.01 Percent Offset)	Roller Load (Pounds)	Material
30,000	18,500	--
35,000	20,000	MIL-S-23284, class 4
40,000	21,500	--
45,000	26,500	MIL-S-23284, class 3
50,000	30,000	--
55,000	32,750	MIL-S-23284, class 2
60,000	36,000	--
65,000	40,000	--
70,000	43,400	--
75,000	46,000	MIL-S-23284, class 1, class 5
80,000	49,000	--
85,000	52,000	--
90,000	55,000	MIL-S-23284, class 6

NOTE:

1/ The roller load on the shaft shall be increased slowly from zero to the load specified in this table starting 2 inches ahead of the section to be rolled and decreased gradually from the load specified in this table to zero over a distance of 2 inches following the section to be rolled.

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5.8.2.8 Inspection. Upon completion of cold rolling and machining the keyways, the rolled area of the shaft shall be inspected by magnetic particle inspection as specified in 5.3.2.

5.8.2.9 Final finishing. The amount of metal removed for final finishing after cold rolling shall be kept to a minimum and shall be not greater than 0.005 inch on the radius.

5.9 Shot peening requirements. Surfaces requiring shot peening shall be magnetic particle or liquid penetrant inspected (see 5.3.2) before processing. Areas of the shaft or stock that have been previously shot peened shall be re-shot peened after stress relief, if weld repaired. Shot peening shall be performed in accordance with SAE AMS2430 or SAE AMS2590, as modified herein. The shot peening intensity and medium shall be as specified in 5.9.1 and 5.9.2, respectively, for given material types, unless otherwise specified on the part drawing or in the contract.

5.9.1 Peening intensity. The shot peening intensity of steel shafts shall be 0.012 to 0.016A. For shafts overlaid with corrosion-resistant cladding (see table I), the shot peening intensity shall be 0.008 to 0.012A.

5.9.2 Shot medium. For steel shafts, the shot medium shall conform to SAE AMS2431/1, type ASR 330 cast steel shot. For shafts overlaid with corrosion-resistant cladding (see table I), the shot medium shall conform to SAE AMS2431/7, type AZB 850 ceramic shot.

5.9.3 Coverage. Shot peened surfaces shall receive full coverage in accordance with SAE J2277.

5.9.4 Flange fillets. Shot peening of flange fillets shall include the entire curvature of the fillet and entire circumference. Repair certification records shall identify that the flange fillet(s) has/have been shot peened. The shaft shall also be die-stamped in the area of other identifying information to indicate that the fillet(s) has/have been shot peened, including the corresponding date and performing facility. If the shaft is flanged on both ends but only one fillet is shot peened, the die stamping shall clearly mark either “FWD fillet” or “AFT fillet”, as applicable. If there is not documentation or clear evidence that a flange fillet has been shot peened during a prior refurbishment period, the fillet shall be shot peened.

5.9.5 Final finishing. The amount of metal removed for final finishing after shot peening shall be kept to a minimum and shall be not greater than 0.002 inch on the radius. Material removal after shot peening shall be recorded for information.

5.9.6 Inspection. If material is removed after shot peening (see 5.9.5), the affected surface shall be magnetic particle or liquid penetrant inspected (see 5.3.2).

5.10 Repair facility approval. Facilities that repair shafts and stocks for the U.S. Navy to the requirements of this standard shall have NAVSEA approval of their ability for compliance. NAVSEA approval is required prior to acceptance of work. NAVSEA approval shall require an audit of the repair facility’s documentation and capabilities to ensure the ability of the repair facility to meet the requirements of this standard.

5.10.1 Requirements for NAVSEA approval of a shafting repair facility.

a. The facility shall have qualified welding procedures to the requirements of NAVSEA S9074-AQ-GIB-010/248 and this standard and maintain the original procedure qualification records.

b. The repair facility shall have welders qualified to the requirements of NAVSEA S9074-AQ-GIB-010/248 and this standard and maintain welder qualification to the requirements of NAVSEA S9074-AQ-GIB-010/248 and this standard.

c. The repair facility shall be physically able to handle shafting without damage.

d. The facility shall have the capability to perform and document a uniform controlled stress relief of shafting.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This standard is intended for use in repair welding, weld cladding, straightening, cold rolling, workmanship, inspection, and record keeping for main propulsion shafting, rudder stocks, and diving plane stocks for ships of the United States Navy. This standard may be used as a guide for repair of other shafting that is not included herein.

6.1.1 Backwards compatibility. Prior welding procedure qualifications may not need to be repeated (see 5.1.1). This revision should be used for new repairs; previous repairs made to prior revisions are acceptable.

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6.2 Subject term (key word) listing.

Gas metal arc
Heat-affected zone
Liquid penetrant inspection
Magnetic particle inspection
Nondestructive testing
Peening
Shielded metal arc
Stress relief
Submerged arc
Ultrasonic inspection

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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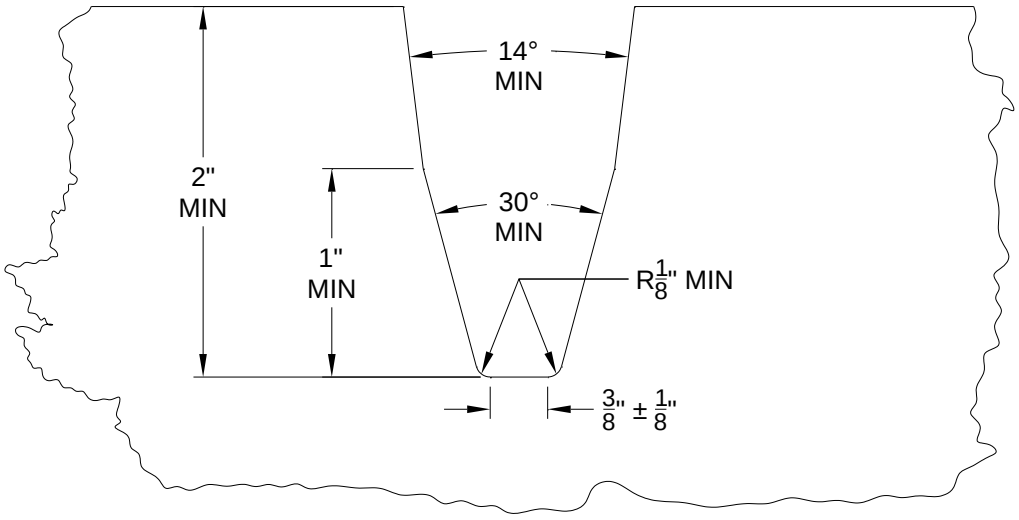


FIGURE 1. SMAW solid shaft joint design.

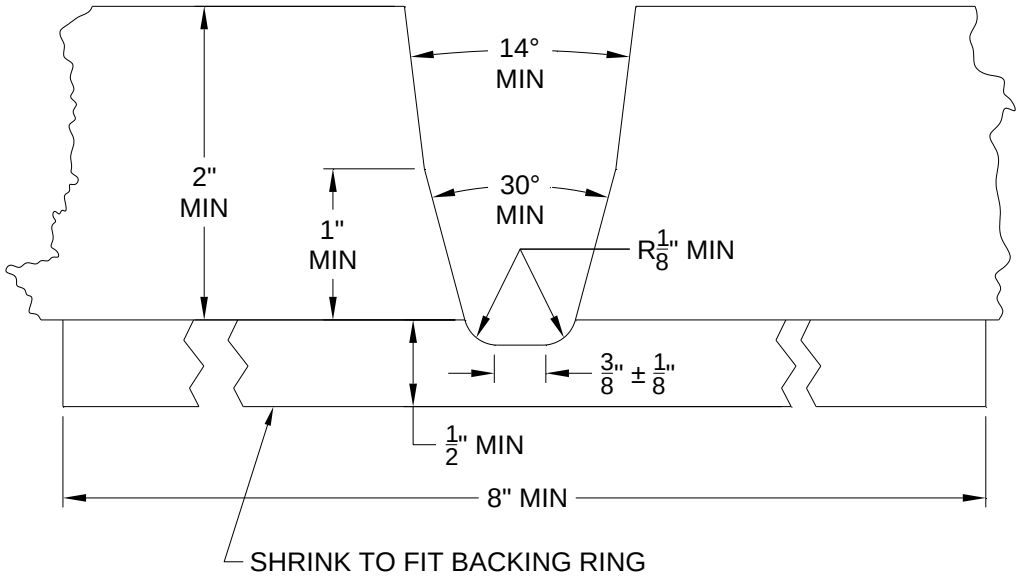


FIGURE 2. SMAW hollow shaft joint design.

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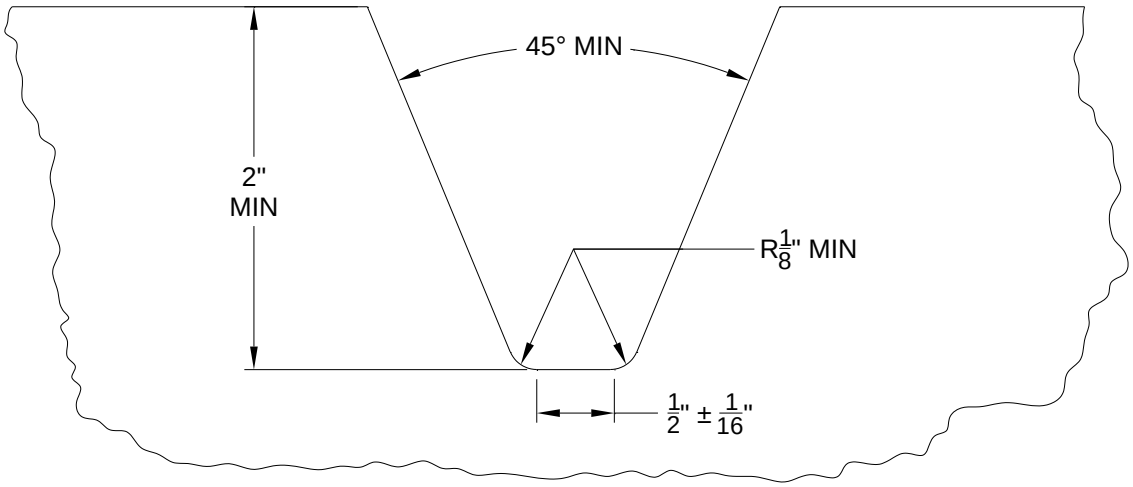


FIGURE 3. GMAW solid shaft joint design.

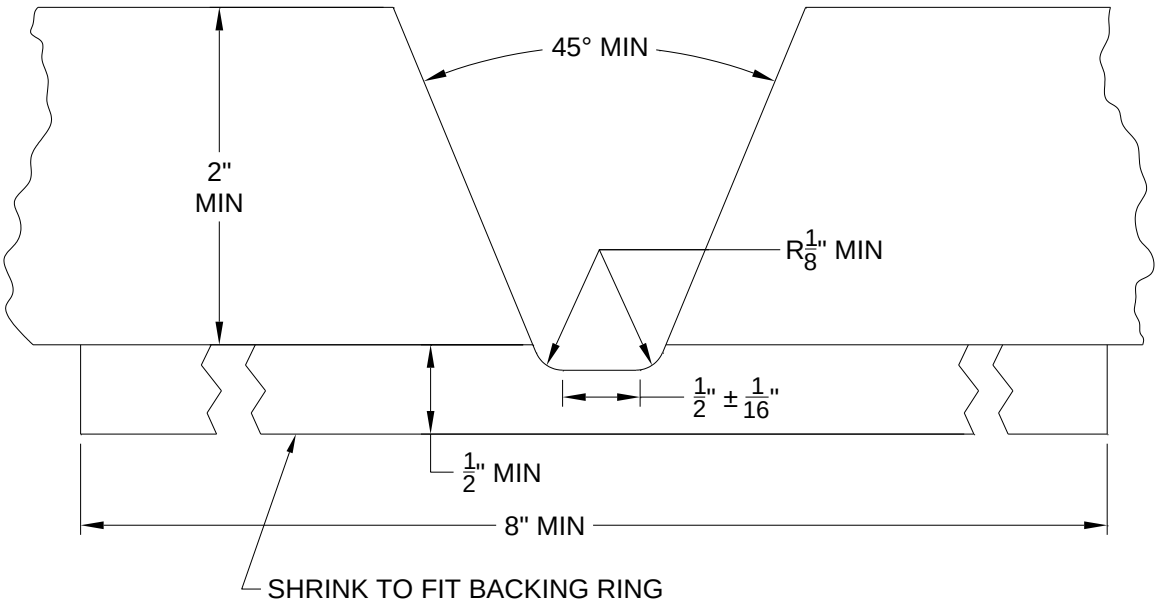


FIGURE 4. GMAW hollow shaft joint design.

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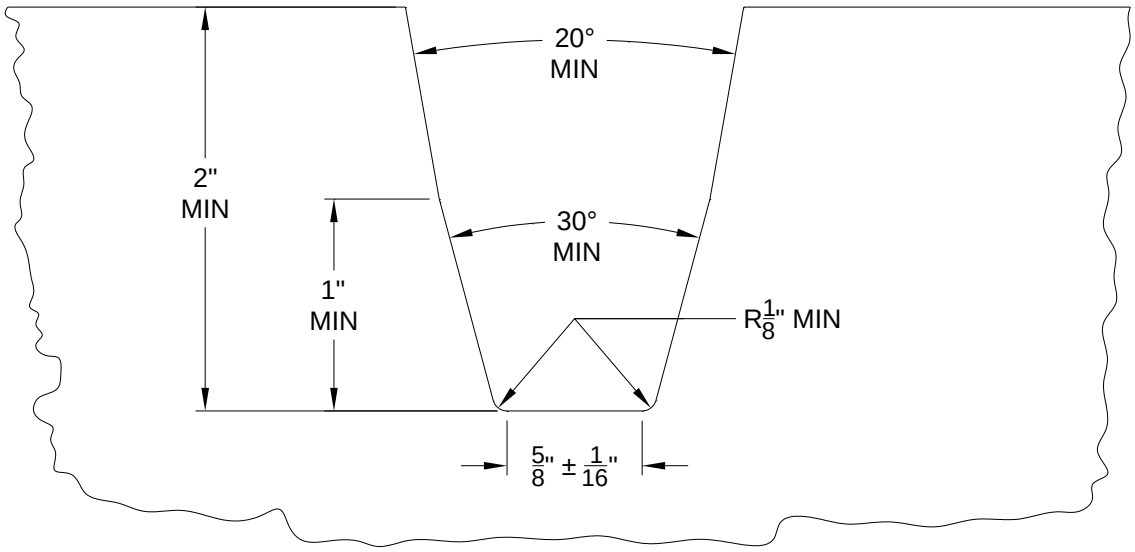


FIGURE 5. SAW solid shaft joint design.

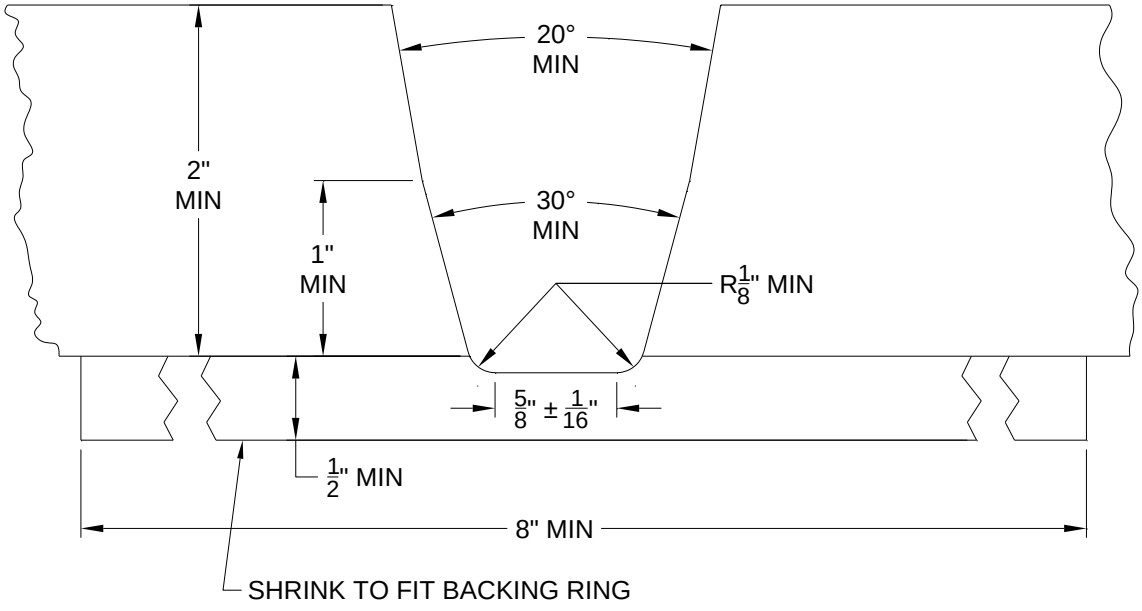


FIGURE 6. SAW hollow shaft joint design.

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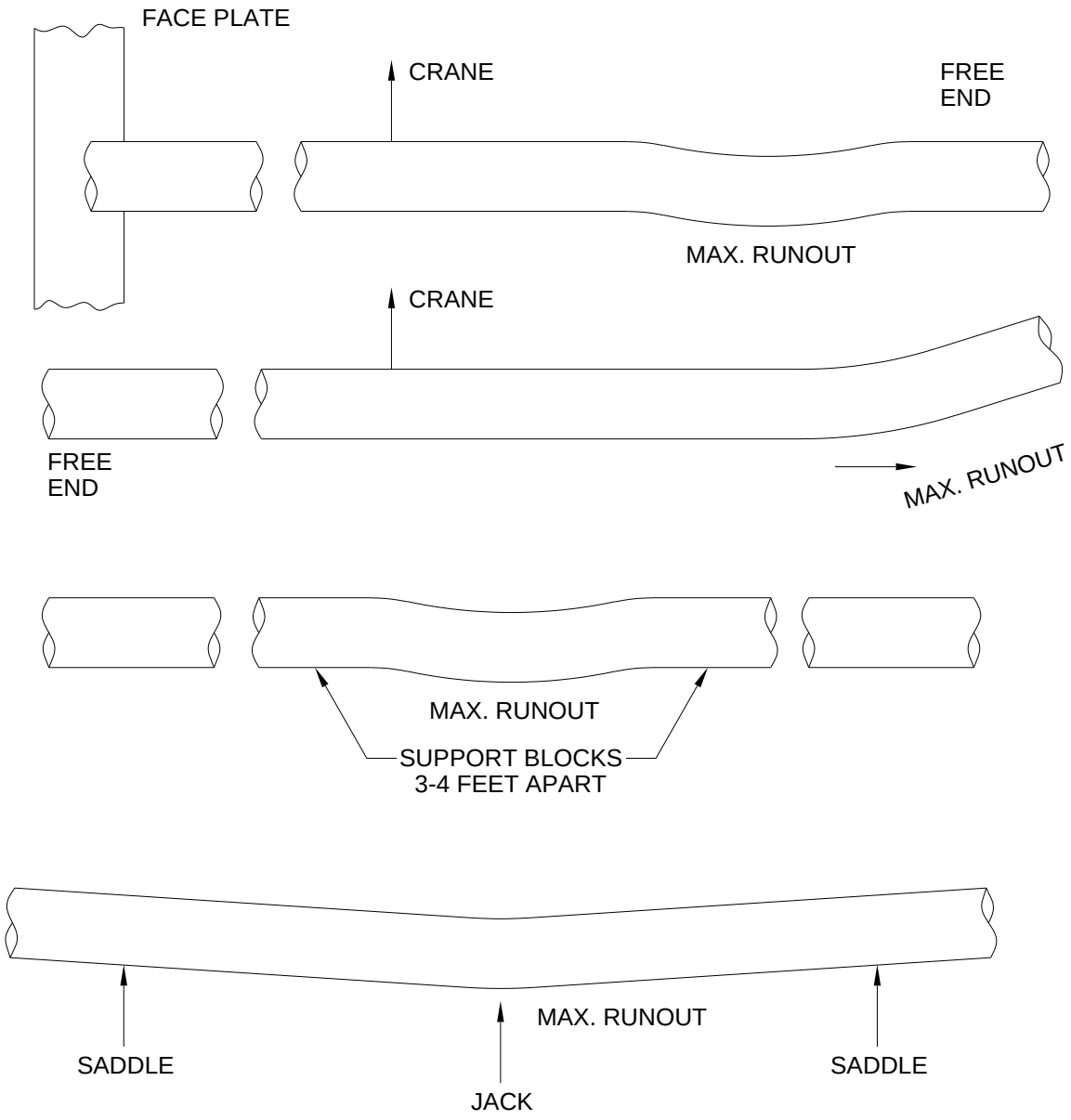


FIGURE 7. Methods of supporting shaft for straightening by vibration.

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CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 2010-2024-001)

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